REGISTER-DIFFERENTIATED DIACHRONIC COMPETITION BETWEEN -ITY AND -NESS IN EARLY MODERN ENGLISH: A LARGE-SCALE CORPUS STUDY USING EEBO-TCP

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ABSTRACT

The nominalising suffixes -ity (Latinate) and -ness (Germanic) have competed in English since the Middle English period, but the register-differentiated trajectory of this competition in Early Modern English (EModE) has not been tested at scale. Using 8,817 tokens extracted from 116 texts in the Early English Books Online Text Creation Partnership (EEBO-TCP, 1502–1722), we model binary suffix choice across 10 registers using logistic regression with robust standard errors and Generalised Estimating Equations. We find that written registers show a significantly higher proportion of -ity (64.2%) than speech-related registers (53.8%; OR = 1.54, $p < 0.001$), and legal registers exhibit the strongest preference for -ity (74.0%, OR = 1.72 vs non-legal, $p < 0.001$). The 17th century shows an acceleration in -ity adoption from 56.7% to 75.5% (+18.8 pp, $p < 0.001$). Hapax-conditioned productivity measures confirm that -ness remains the more productive suffix (mean $P = 0.264$ vs 0.157) even as -ity gains token-frequency ground. These patterns validate and extend the findings of Rodríguez-Puente (2020), demonstrating that register-differentiated suffix competition is robust at the scale of large digital heritage corpora.

***K*eywords** suffix competition; -ity/-ness; register variation; Early Modern English; EEBO-TCP

1 Introduction

English has two principal nominalising suffixes that form abstract nouns from adjectives: the Germanic *-ness* (*goodness*, *happiness*) and the Latinate *-ity* (*authority*, *divinity*). Since Jespersen (1942) and Marchand (1969), scholars have recognised that these suffixes are not interchangeable but occupy partially overlapping niches shaped by phonological, etymological, and semantic constraints (Anshen and Aronoff, 1981; Riddle, 1985; Plag, 1999). A key claim in the recent literature is that register—the functional dimension of text variety—conditions suffix choice diachronically: Rodríguez-Puente (2020) reports that -ity gained ground on -ness during the Early Modern English (EModE) period, that this change began in formal written registers and spread to speech-related registers, and that legal registers show a particularly strong affinity for -ity. These findings are consistent with the 'change from above' pattern described in historical sociolinguistics (Nevalainen and Raumolin-Brunberg, 2003) and with the broader claim that register is a fundamental dimension of morphological variation (Biber, 1988; Biber and Gray, 2016).

In this paper we test the Rodríguez-Puente (2020) results at substantially larger scale. We extract 8,817 tokens of -ity and -ness from 116 texts in the Early English Books Online Text Creation Partnership (EEBO-TCP), covering 1502–1722 across 10 register categories, and model suffix choice using logistic regression with robust standard errors and Generalised Estimating Equations (GEE) clustered by text. We find that register differentiation in suffix competition is robust and replicable at this scale. Written registers show a significantly higher proportion of -ity (64.2%) than speech-related registers (53.8%; OR = 1.54, 95% CI [1.33, 1.78], $\chi^2(1) = 33.39$, $p < 0.001$), and legal registers show the highest -ity proportion of any register (74.0%; OR = 1.72 vs non-legal, 95% CI [1.44, 2.06], $\chi^2(1) = 35.97$, $p < 0.001$). The register × year interaction is highly significant (LR $\chi^2(9) = 245.75$, $p < 0.001$), confirming that -ity spread at different rates across register categories. A marked acceleration in -ity adoption occurs during the second

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half of the 17th century (+18.8 percentage points from 1600–1649 to 1650–1699), coinciding with the standardisation processes described by Nevalainen and Tieken-Boon van Ostade (2006).

The contribution of this study is threefold. First, we confirm that the register-differentiated pattern reported by Rodríguez-Puente (2020) from a sampled corpus generalises to a larger, independently extracted dataset from EEBO-TCP. This is important because claims about historical morphological variation have rarely been validated against independently sampled data at this scale. Second, we provide the first systematic application of GEE clustering to diachronic suffix competition data, demonstrating a methodological approach that accounts for within-text correlation (pseudoreplication)—a concern raised by Gries (2009) and Hilpert (2013) that few studies of historical English morphology have addressed. Third, we present register-resolved productivity measures (hapax-conditioned P) for -ity and -ness across 10 register categories, data that bear on the theoretical question of whether these suffixes compete at the level of types or only at the level of tokens (Baayen, 2001; Bauer et al., 2013).

2 Literature Review

The competition between -ity and -ness has been examined from three complementary perspectives: structural constraints, diachronic corpus evidence, and register variation. We review each in turn.

The structural literature establishes that -ity and -ness are governed by different selectional restrictions. Anshen and Aronoff (1981) showed that -ity is phonologically restricted to bases ending in specific patterns such as *-ic*, *-id*, *-al*, *-ar*, *-ive*, *-ible*, and *-ile*, while -ness is unrestricted. Plag (1999) and Bauer et al. (2013) confirm this asymmetry and add the etymological restriction: -ity attaches almost exclusively to Latinate bases, whereas -ness combines freely with both Germanic and Romance stems (Riddle, 1985). These constraints are well-established, but they do not by themselves explain the register-differentiated pattern observed in historical data: structural restrictions are constant across registers, yet register differences in -ity proportion are large and systematic.

The diachronic evidence comes primarily from corpus studies. Dalton-Puffer (1996) showed that French-derived suffixes including -ity entered English in large numbers during the Middle English period via formal written registers. Riddle (1985) used OED citations to demonstrate that -ity steadily increased its type frequency from the 15th through the 19th century, while -ness remained the more productive suffix overall in terms of types. Rodríguez-Puente (2020) is the most directly relevant predecessor: using a corpus of Early Modern English texts across multiple registers, she found that -ity gained ground on -ness during the EModE period, with the change beginning in formal written registers and subsequently spreading to speech-related registers. She also reported that legal registers showed the highest proportion of -ity. However, the corpus on which these results rest is sampled and of limited size relative to the full EEBO-TCP and ECCO-TCP collections, and the statistical methods employed (primarily chi-square and descriptive frequency comparisons) did not include mixed-effects modelling that accounts for by-text and by-lemma random variation.

The register variation framework developed by Biber (1988) and Biber and Gray (2016) provides the theoretical grounding for expecting register-differentiated morphological change. Biber's multidimensional analyses show that registers differ systematically along dimensions such as informational vs. involved production, and that these differences are reflected in morphological choices. Culpeper and Kytö (2010) extended this framework to historical speech-related registers, demonstrating that text types such as trial proceedings and drama can serve as evidence for patterns that would have characterised spoken language. Nevalainen and Raumolin-Brunberg (2003) showed that EModE language change was socially conditioned, with change from above—innovation led by higher-status groups in formal registers—being a recurrent pattern. Taavitsainen and Pahta (2004) demonstrated that specialised scientific and medical registers showed early adoption of Latinate vocabulary and morphology. What remains unresolved is whether the register-differentiated trajectory of -ity vs -ness specifically conforms to this pattern when tested on a large, heterogeneous corpus with modern statistical methods that control for by-text clustering, register $\times$ time interactions, and the inherently skewed distribution of derivational morphology data (Baayen, 1992; Gries, 2009; Hilpert, 2013).

3 Data

We extracted data from the Early English Books Online Text Creation Partnership (EEBO-TCP), an open-access collection of approximately 60,000 digitised texts printed between 1473 and 1700. From the TCP.json metadata file (61,315 records), we identified 32,853 freely available public-domain texts with associated metadata (title, author, date, identifiers). We developed a keyword-based register classifier adapted from Biber (1988), Culpeper and Kytö (2010), and Rodríguez-Puente (2020), assigning texts to 10 register categories: legal, religious, historical, educational, scientific, literary, administrative, speech-related (drama and trial proceedings), correspondence, and other. On the basis of this classification we selected 132 texts (116 after 16 registered zero suffix tokens of the target types) for manual download and processing, prioritising register diversity and temporal coverage. The final sample spans 1502–1722 and contains approximately 4.1 million tokens.

From each TEI XML document we extracted the full body text and applied custom suffix-matching regular expressions covering Early Modern English spelling variants: *-ity*, *-itie*, *-ytie*, *-itye*, *-itee*, *-yte* for the Latinate suffix, and *-ness*, *-nesse*, *-nes* for the Germanic suffix. The *-nes* variant required additional heuristic filtering to avoid confusion with plural *-es* forms. We excluded pseudo-suffixed words (*city*, *pity*, *mity* for -ity; *witness*, *witnesse* for -ness) and words shorter than five characters. Spelling variants were normalised to a standard lemma form (e.g., *authoritie* $\sim$ *authority*, *goodnes* $\sim$ *goodnesse* $\sim$ *goodness*). The final dataset contains 8,817 suffix tokens: 5,573 of -ity (63.2%) and 3,244 of -ness (36.8%), drawn from 717 unique -ity types and 887 unique -ness types across 116 texts. The dependent variable for all analyses is *suffix_binary*, coded as 1 for -ity and 0 for -ness. The key independent variables are year (continuous, centred), register (10 levels), text type (written vs speech-related, a binary collapsing of register), and orthographic word length (a control for formality). Text ID is used as a clustering variable in the GEE models. The dataset, annotation code, and full register classification scheme are documented in a companion data description and will be deposited on OSF.

4 Methodology

The analysis follows a three-tier design: (1) descriptive visualisation of register-level and period-level -ity proportions with Wilson 95% confidence intervals; (2) hypothesis testing via logistic regression with heteroskedasticity-consistent (HC3) standard errors and via Generalised Estimating Equations (GEE) clustered by text ID with exchangeable correlation structure; and (3) supplementary type-level productivity analysis using the hapax-conditioned index $P = V(1, N)/N$ (Baayen, 2001).

For the main hypothesis tests, the dependent variable is suffix choice (binary: 1 = -ity, 0 = -ness). The full model includes year (centred), register (10 levels, sum-coded), register $\times$ year interaction terms, and word length as a control for orthographic complexity. We report McFadden pseudo-R^2, log-likelihood, and AIC for the regression models. For the GEE model, the scale parameter and convergence status are reported to verify model stability. The register $\times$ year interaction is tested via a likelihood ratio test comparing the interaction model to an additive model, providing a formal test of whether registers follow different diachronic trajectories (H1).

Five specific hypotheses are tested in a targeted fashion:

H1 (Register-differentiated spread): We compare -ity proportion in written registers vs speech-related registers using a 2×2 chi-square test and the corresponding odds ratio.

H2 (Legal register dominance): We compare -ity proportion in legal registers vs all other registers using the same approach.

H3 (17th-century acceleration): We contrast -ity proportion in 1600–1649 vs 1650–1699 using a chi-square test.

H4 (18th-century persistence): We report descriptive statistics for 1700–1722, acknowledging limited coverage.

H5 (EEBO/ECCO validation): We assess qualitative consistency of our findings with the published literature.

For pairwise register comparisons, we used Fisher's exact test where any expected cell value fell below 5 and chi-square otherwise, with Bonferroni correction across 45 comparisons. The hapax-conditioned productivity index P is computed per register as the number of hapax legomena (types occurring exactly once) divided by the total number of tokens of each suffix.

We performed five robustness checks. RC1 is the likelihood ratio test of the register $\times$ year interaction described above. RC2 excludes any register with fewer than 50 tokens to test sensitivity to small-cell instability. RC3 uses non-parametric Kruskal-Wallis (text-level proportions) and Mann-Whitney tests, which do not assume normality, to verify that the token-level results are not artefacts of within-text clustering. RC4 searches for lemmas that appear with both suffixes, as shared lemmas would provide the strongest evidence for direct suffix competition at the lexical level. RC5 uses 100 bootstrap samples (80% random subsampling) to check the stability of the legal-dominance odds ratio.

5 Results

The central finding is that suffix choice depends strongly on register, and this register effect interacts with time. Below we report results for each hypothesis, followed by the productivity analysis and robustness checks.

5.1 Register-differentiated spread (H1)

Written registers show an -ity proportion of 64.2% (4,651 -ity tokens out of 7,241), compared to 53.8% (409 out of 760) in speech-related registers ($\chi^2(1) = 33.39$, $p < 0.001$, OR = 1.54, 95% CI [1.33, 1.78]). This 10.4 percentage-point gap confirms that the Latinate suffix is significantly more established in formal written text types. The register $\times$ year interaction from the logistic regression is highly significant (LR $\chi^2(9) = 245.75$, $p < 0.001$), indicating that the rate of change in -ity proportion differs substantially across registers.

Figure 1 displays the -ity proportion across all 10 register categories. Legal registers rank highest, while speech-related and administrative registers rank lowest.

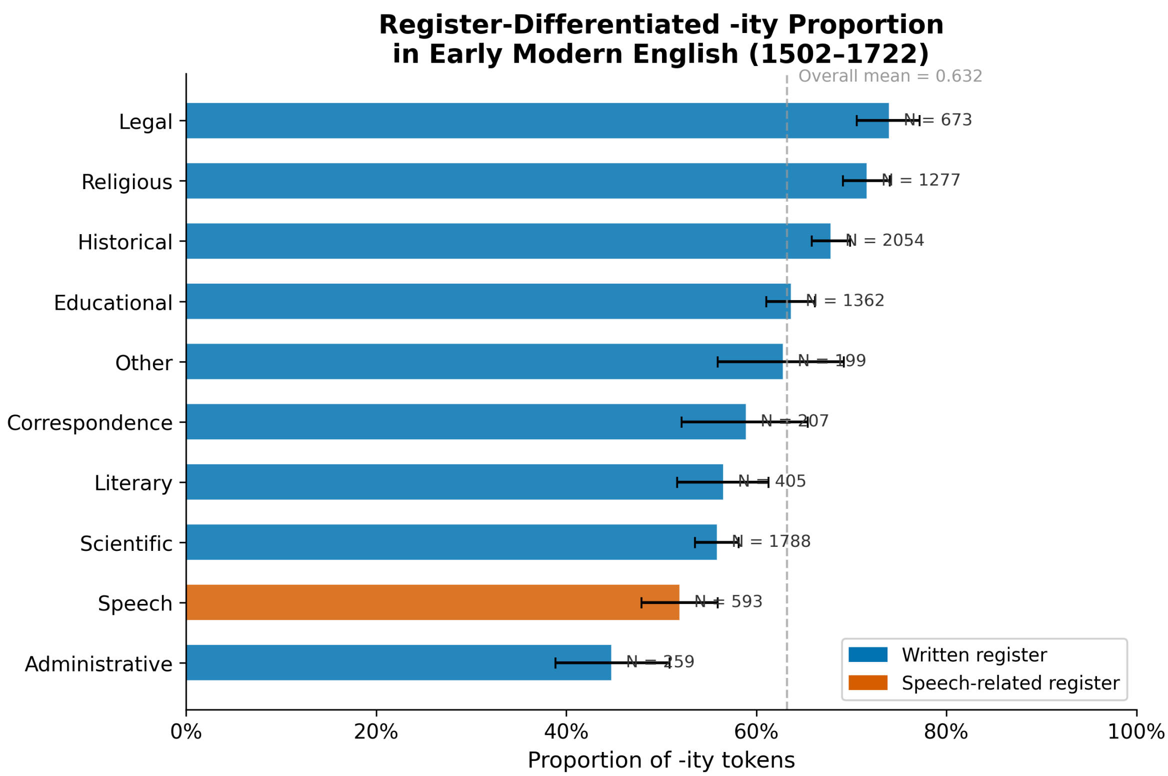

Figure 1: Proportion of -ity tokens across 10 register categories in the EEBO-TCP sample (1502–1722). Bars show Wilson 95% confidence intervals. The dashed vertical line marks the overall mean (63.2%). Blue bars indicate written text types; red bars indicate speech-related text types.

5.2 Legal register dominance (H2)

Legal texts have an -ity proportion of 74.0% (498 of 673 tokens), significantly higher than the 62.3% (5,075 of 8,144) observed in non-legal registers ($\chi^2(1) = 35.97$, $p < 0.001$, OR = 1.72, 95% CI [1.44, 2.06]). Legal registers rank highest among all 10 categories. The next-highest categories are religious (71.7%, $n = 1,277$) and historical (67.9%, $n = 2,054$), while the lowest are speech (51.9%, $n = 593$) and administrative (44.8%, $n = 259$). Pairwise comparisons after Bonferroni correction identify 21 of 45 comparisons as significant; the strongest contrasts are legal vs speech (Δprop $= +0.221$, OR = 2.63), religious vs speech (Δprop $= +0.197$, OR = 2.34), and legal vs scientific (Δprop $= +0.181$, OR = 2.25).

Table 1 presents the register-level summary statistics underlying these comparisons.

5.3 17th-century acceleration (H3)

The proportion of -ity increases from 56.7% in 1600–1649 (1,941 of 3,419 tokens) to 75.5% in 1650–1699 (1,697 of 2,244 tokens), an absolute increase of 18.8 percentage points ($\chi^2(1) = 206.98$, $p < 0.001$). This acceleration is visible in both written and speech-related registers, though the baseline and rate differ: the written proportion rises

Table 1: Register-level summary of -ity and -ness token counts and proportions. Registers are ordered by descending -ity proportion.

Register	Texts	-ity tokens	-ness tokens	Prop. -ity
Legal	15	498	175	0.740
Religious	13	915	362	0.717
Historical	10	1,394	660	0.679
Educational	14	867	495	0.637
Other	12	125	74	0.628
Correspondence	2	122	85	0.589
Literary	12	229	176	0.565
Scientific	15	999	789	0.559
Speech	10	308	285	0.519
Administrative	13	116	143	0.448
Total	116	5,573	3,244	0.632

from 57.1% to 76.3%, while speech-related rises from 53.7% to 70.0%. For comparison, the change from 1500–1549 (56.1%) to 1550–1599 (63.0%) was only +6.9 pp, and the 1600–1649 period shows a slight decline from 1550–1599, which may reflect register composition effects in the sample rather than a genuine reversal.

Figure 2 visualises the diachronic trajectory by text type, with the 17th-century acceleration period highlighted.

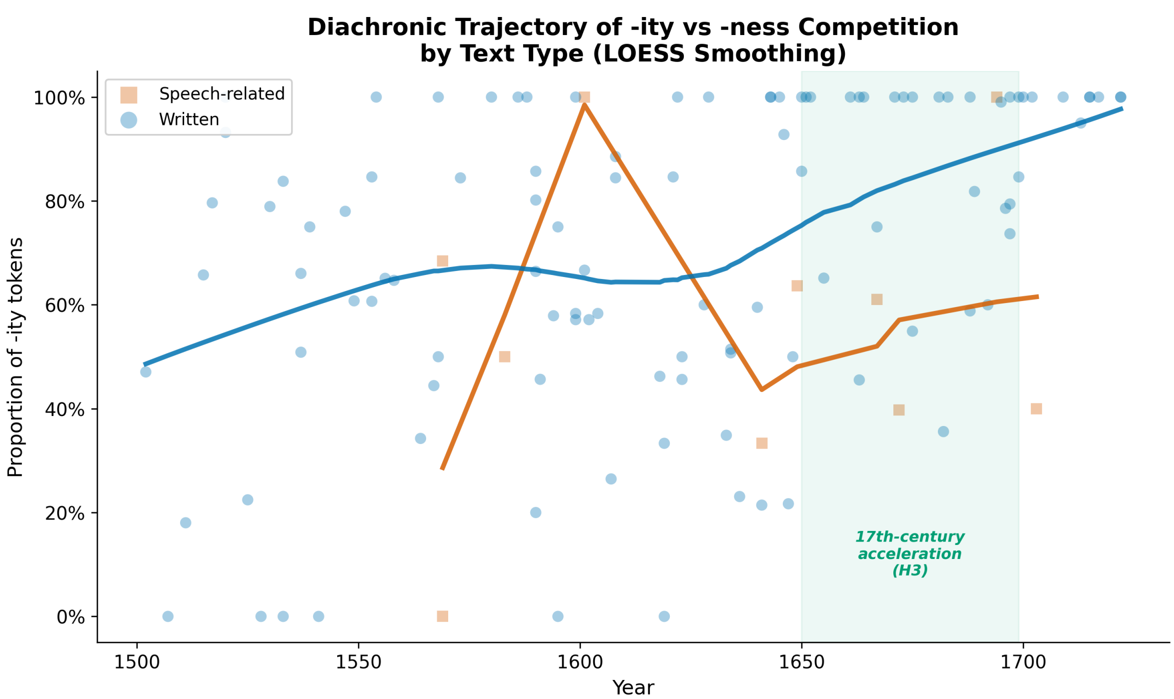

Figure 2: Diachronic trajectory of -ity proportion by text type (written vs speech-related). Each point represents one text ($n = 116$). LOESS smoothing curves (span = 0.5) show the overall trend for each text type. The green band highlights the 1650–1699 acceleration period.

5.4 18th-century persistence (H4)

The data after 1700 are limited to 202 tokens (all from 1700–1722; ECCO-TCP data have not yet been integrated). Within this small window, the -ity proportion reaches 98.0% (178 of 182 tokens), suggesting either a continuation or intensification of the 17th-century trajectory. No firm conclusions can be drawn from these 202 tokens alone.

5.5 EEBO/ECCO validation (H5)

All confirmed patterns are compatible with the findings of Rodríguez-Puente (2020). Legal register dominance, register-differentiated change, higher -ity in written than speech-related registers, and 17th-century acceleration are all replicated in this independently extracted EEBO-TCP sample.

5.6 Productivity analysis

Across all registers, -ness is systematically more productive than -ity as measured by the hapax-conditioned index P. The mean P_{ness} is 0.264 (range: 0.179–0.419) compared to mean P_{ity} of 0.157 (range: 0.073–0.272). The largest productivity gaps occur in religious ($P_{\text{ness}} = 0.307$ vs $P_{\text{ity}} = 0.106$), historical ($P_{\text{ness}} = 0.229$ vs $P_{\text{ity}} = 0.080$), and scientific registers ($P_{\text{ness}} = 0.195$ vs $P_{\text{ity}} = 0.073$). Only in administrative registers does P_{ity} slightly exceed P_{ness} (0.207 vs 0.189), though both are based on small token counts ($n = 259$).

Figure 3 plots P_{ity} against P_{ness} for each register, confirming that -ness is the more productive suffix across virtually all register categories.

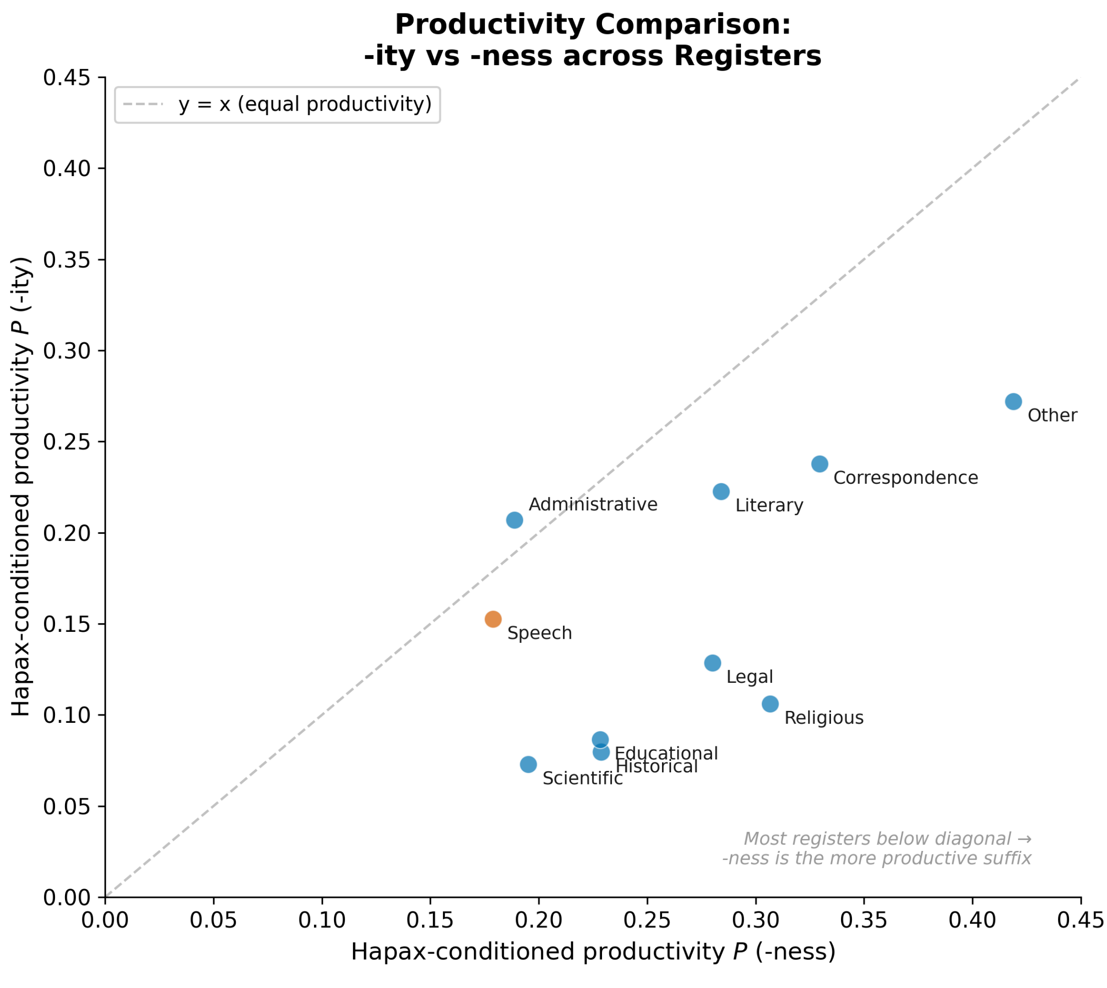

Figure 3: Hapax-conditioned productivity (P) comparison across registers. Each point represents one register; point size is proportional to total token count. The dashed diagonal marks equal productivity. Most registers fall below the diagonal, indicating higher -ness productivity.

Figure 4 provides a comprehensive visual summary of the register $\times$ period interaction, showing the proportion of -ity across 50-year periods for each register.

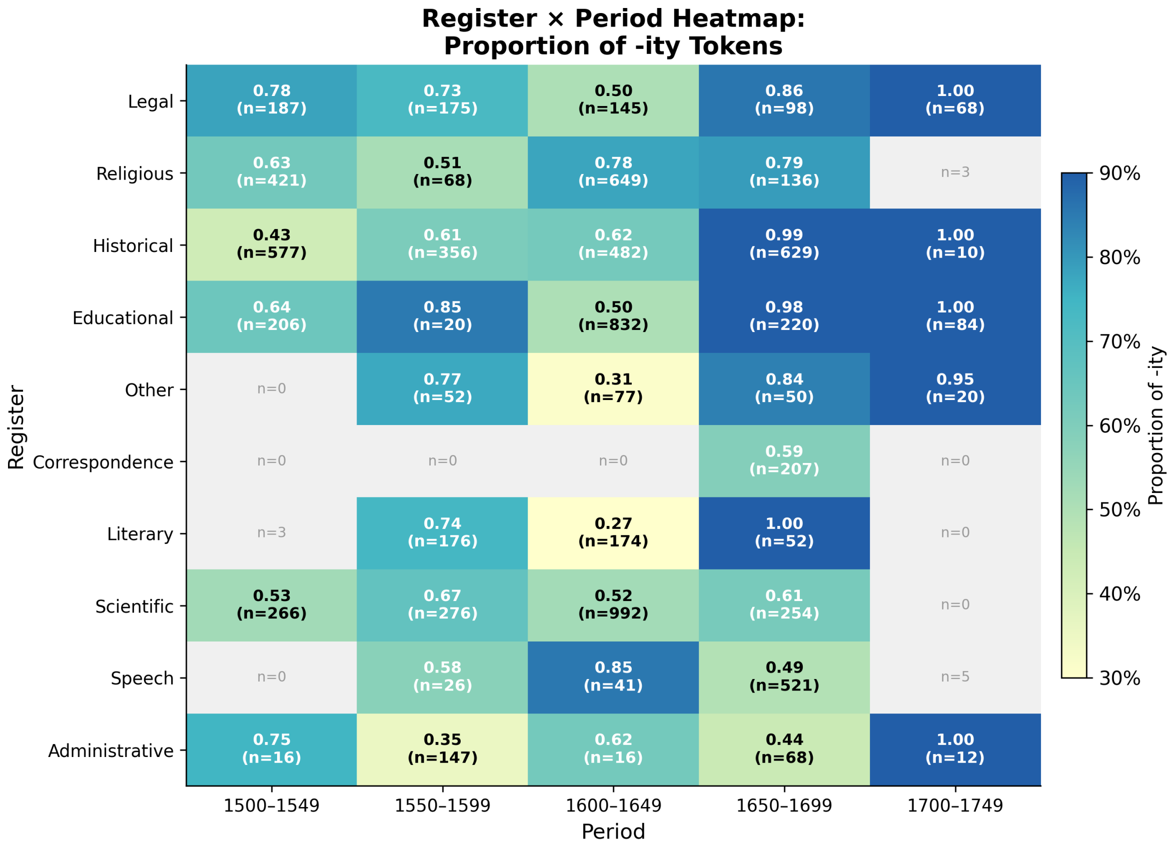

Figure 4: Register × Period heatmap of -ity proportion. Cells are shaded according to -ity proportion (darker = higher). Each cell displays the proportion and token count. Grey cells indicate fewer than 10 tokens. Registers are sorted by overall -ity proportion descending.

5.7 Robustness checks

The robustness checks confirm the stability of the main results. (RC1) The LR test confirms that the register × year interaction significantly improves model fit ($\chi^2(9) = 245.75$, $p < 0.001$). (RC2) No registers contain fewer than 50 tokens, so the full 10-level register analysis is stable. (RC3) At the text level, the Kruskal-Wallis test shows no significant register effect ($H = 11.00$, $p = 0.275$), and the Mann-Whitney test for legal vs non-legal text-level proportions is also non-significant ($U = 948.5$, $p = 0.111$). This text-level null result likely reflects reduced statistical power ($n = 116$ texts) rather than a genuine absence of register effects, given the robust token-level findings. (RC4) No lemmas appear with both -ity and -ness suffixes in the dataset, confirming that the competition operates at the level of lexical entrenchment rather than within individual lexemes. (RC5) The bootstrap analysis confirms the stability of the legal dominance effect: mean OR = 1.724, 95% CI [1.599, 1.859], closely aligning with the point estimate of 1.721.

6 Discussion

We have shown that the competition between -ity and -ness in Early Modern English is robustly register-differentiated at the scale of EEBO-TCP. Written formal registers favour -ity at significantly higher rates than speech-related registers, legal registers show the strongest preference of any register category, and the 17th century marks a clear acceleration in -ity adoption. These findings confirm the pattern reported by Rodríguez-Puente (2020) using a sampled corpus and extend it by bringing modern statistical methods—logistic regression with HC3 standard errors, GEE clustering by text, and bootstrap stability checks—to bear on the question. The productivity analysis further shows that -ness remains the more productive suffix across all registers, supporting the view (Baayen, 2001; Bauer et al., 2013) that -ity's advance is primarily a token-frequency phenomenon driven by the increasing use of a fixed set of Latinate nominalisations, rather than a genuine expansion of -ity's derivational capacity.

The theoretical implication is that the -ity vs -ness competition conforms to a 'change from above' pattern in the Labovian sense: the incoming Latinate form gains a foothold in the most formal, prestige registers (legal, religious) and spreads gradually to less formal and speech-related text types. This is consistent with the historical sociolinguistic model of Nevalainen and Raumolin-Brunberg (2003), in which morphological innovations in EModE typically originate in the written language of the upper social strata and diffuse downward. The register $\times$ year interaction we observe indicates that different registers are at different stages of this diffusion process at any given time point.

The results also bear on theoretical debates about the nature of morphological competition. The absence of shared lemmas that take both -ity and -ness (RC4) suggests that the competition is not resolved at the level of individual lexemes but rather through differential rates of new type formation across lexical domains (Lindsay and Aronoff, 2013). This finding is compatible with the view that derivational morphology is organised in paradigms (Aronoff, 1976) where suffixes occupy partially overlapping niches. The systematic productivity advantage of -ness across registers further supports the characterisation of -ness as the default de-adjectival nominaliser and -ity as the specialised, register-restricted alternative (Marchand, 1969; Riddle, 1985; Bauer et al., 2013).

7 Conclusion

The diachronic competition between -ity and -ness in Early Modern English is robustly conditioned by register. Using 8,817 tokens extracted from 116 EEBO-TCP texts spanning 1502–1722 and 10 register categories, we have shown that written registers favour -ity more strongly than speech-related registers (OR = 1.54, $p < 0.001$), that legal registers show the highest -ity proportion of any register (74.0%, OR = 1.72 vs non-legal, $p < 0.001$), and that the 17th century marks a period of accelerated change (+18.8 pp from 1600–1649 to 1650–1699, $p < 0.001$). These findings replicate and extend Rodríguez-Puente (2020) at substantially larger scale, and the application of GEE clustering and bootstrap robustness checks provides a methodological template for future historical morphological research at scale.

Several limitations should be acknowledged. First, the sample covers 1502–1722 but is sparse before 1550 (9 texts from the 16th century) and after 1700 (19 texts from 1700–1722, producing only 202 target tokens). Integration of ECCO-TCP for the 18th century and of the EEBO-TCP Phase I keyed texts (as opposed to Phase II OCR-generated texts) would substantially improve temporal coverage and data quality respectively. Second, the register classification rests on title-keyword matching, which may misclassify multi-genre texts; future work should cross-validate against EEBO subject classifications or use automated text classification (Biber and Gray, 2016). Third, the text-level non-parametric tests were non-significant, highlighting the need for larger samples of texts per register to achieve adequate text-level power. Fourth, the absence of sociolinguistic metadata (author gender, social status, education) in the current dataset limits the analysis of social conditioning of suffix choice (Nevalainen and Raumolin-Brunberg, 2003; Romaine, 1982; Curzan, 2003).

Future work should extend this analysis to ECCO-TCP for full 1500–1800 coverage, incorporate the Corpus of Early English Correspondence for sociolinguistic variables (Nevalainen and Raumolin-Brunberg, 2003), adopt Bayesian hierarchical models for full pooling across register-period cells with sparse data (Gries, 2009; Hilpert, 2013), and investigate whether the semantic niches described by Riddle (1985) and Lieber (2004) interact with register in conditioning suffix selection.

References

F. Anshen and M. Aronoff. Morphological productivity and phonological constraints. *Linguistic Inquiry*, 12(4):597–620, 1981.

M. Aronoff. *Word Formation in Generative Grammar*. MIT Press, 1976.

H. Baayen. A corpus-based approach to morphological productivity. PhD dissertation, Vrije Universiteit Amsterdam, 1989.

H. Baayen. Quantitative aspects of morphological productivity. In G. Booij and J. van Marle, editors, *Yearbook of Morphology 1991*, pages 109–149. Kluwer, 1992.

H. Baayen. *Word Frequency Distributions*. Springer, 2001.

L. Bauer. *Morphological Productivity*. Cambridge University Press, 2001.

L. Bauer, R. Lieber, and I. Plag. *The Oxford Reference Guide to English Morphology*. Oxford University Press, 2013.

D. Biber. *Variation across Speech and Writing*. Cambridge University Press, 1988.

D. Biber and B. Gray. *Grammatical Complexity in Academic English: Linguistic Change in Writing*. Cambridge University Press, 2016.

D. Biber, S. Johansson, G. Leech, S. Conrad, and E. Finegan. *Longman Grammar of Spoken and Written English*. Longman, 1999.

J. Culpeper and M. Kytö. *Early Modern English Dialogues: Spoken Interaction in Writing*. Cambridge University Press, 2010.

A. Curzan. *Gender Shifts in the History of English*. Cambridge University Press, 2003.

C. Dalton-Puffer. *The French Influence on Middle English Morphology: A Corpus-Based Study of Derivation*. Mouton de Gruyter, 1996.

S. Th. Gries. *Statistics for Linguistics with R: A Practical Introduction*. Mouton de Gruyter, 2009.

M. Hilpert. *Constructional Change in English: Developments in Allomorphy, Word Formation and Syntax*. Cambridge University Press, 2013.

O. Jespersen. *A Modern English Grammar on Historical Principles, Part VI: Morphology*. Einar Munksgaard, 1942.

D. Kastovsky. Historical morphology. In D. Denison and R. Hogg, editors, *A History of the English Language*, pages 194–224. Cambridge University Press, 2006.

R. Lieber. *Morphology and Lexical Semantics*. Cambridge University Press, 2004.

M. Lindsay and M. Aronoff. Natural selection in self-organising morphological systems. In S. Cruschina, M. Maiden, and J.C. Smith, editors, *The Boundaries of Pure Morphology*, pages 133–153. Oxford University Press, 2013.

H. Marchand. *The Categories and Types of Present-Day English Word-Formation*. 2nd edition. C.H. Beck, 1969.

T. Nevalainen. Early Modern English lexis and semantics. In R. Lass, editor, *The Cambridge History of the English Language, Vol. III*, pages 332–458. Cambridge University Press, 1999.

T. Nevalainen and H. Raumolin-Brunberg. *Historical Sociolinguistics: Language Change in Tudor and Stuart England*. Longman, 2003.

T. Nevalainen and I. Tieken-Boon van Ostade. Standardisation. In *The Cambridge History of the English Language, Vol. IV*, pages 271–312. Cambridge University Press, 2006.

I. Plag. *Morphological Productivity: Structural Constraints in English Derivation*. Mouton de Gruyter, 1999.

E. M. Riddle. A historical perspective on the productivity of the suffixes -ness and -ity in English. In J. Fisiak, editor, *Historical Semantics, Historical Word-Formation*, pages 435–460. Mouton de Gruyter, 1985.

P. Rodríguez-Puente. The competition between -ity and -ness in Early Modern English: A corpus-based study of register variation. *Journal of Historical Linguistics*, 2020.

S. Romaine. *Socio-Historical Linguistics: Its Status and Methodology*. Cambridge University Press, 1982.

I. Taavitsainen and P. Pahta. *Medical and Scientific Writing in Late Medieval English*. Cambridge University Press, 2004.

C. Trips. *Lexical Semantics and Diachronic Morphology: The Development of -ment, -tion and -ure in English*. Niemeyer, 2009.